5. Use the Porter Stemmer algorithm to perform word stemming on a list of words using python libraries.

Aim: To perform word stemming using the Porter Stemmer algorithm in Python.

Procedure:

1. Install NLTK.
2. Import PorterStemmer.
3. Create a stemmer object.
4. Apply stemming to a list of words.
5. Display the original words and stems.

Code:

```
from nltk.stem import PorterStemmer
ps = PorterStemmer()
words = ["playing", "played", "plays", "studies", "running"]
for word in words:
    print(word, "->", ps.stem(word))
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/5.py"
playing -> play
played -> play
plays -> play
studies -> studi
running -> run
```

Result: The Porter Stemmer algorithm successfully reduced words to their stems.